

Yohan Jun

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Research Areas

Advanced Neuroimaging with MRI	Accelerated MRI, Diffusion MRI, Quantitative MRI
Computational Algorithms for Medical Imaging	Inverse Problem, MR Image Reconstruction, Self-Supervised/Zero-Shot Learning
AI for Automatic Diagnosis of Brain Disorders	Automatic Diagnosis of Brain Tumors Using Deep Learning Algorithms

Education

Ph.D. in Electrical & Electronic Engineering

Seoul, S.Korea

— Yonsei University

Feb. 2022

- Advisor: Dr. Dosik Hwang
- Thesis: "Model-based Deep Learning Reconstruction Methods for Fast Magnetic Resonance Imaging"
- Scholarship: Brain Korea 21 Plus Outstanding Student Fellow Scholarship of Korea Research Foundation
- Award: Best Graduate Student Paper Award

B.S. in Electrical & Electronic Engineering

Seoul, S.Korea

— Yonsei University

Feb. 2016

- Scholarship: National Scholarship for Science & Engineering of Korea Student Aid Foundation

Professional Experience

Assistant Professor, Korea University

Seoul, S.Korea

— Assistant Professor, Korea University College of Medicine, Department of Convergence Medicine

Sep. 2026 - Present

Instructor in Radiology, Harvard Medical School

Boston, US

- Investigator, Martinos Center for Biomedical imaging, Massachusetts General Hospital
- Instructor and Faculty Member in Radiology, Harvard Medical School
- Pediatric Imaging Research Center, Massachusetts General Hospital

Nov. 2024 - Aug. 2026

Research Fellow in Radiology, Harvard Medical School

Boston, US

- Postdoctoral Research Fellow, Martinos Center for Biomedical Imaging, Massachusetts General Hospital
- Advisors: Dr. Berkin Bilgic, Dr. Michael S Gee

Mar. 2022 - Nov. 2024

Research Assistant, Yonsei University

Seoul, S.Korea

- Research Assistant, Medical Artificial Intelligence Lab.
- Advisor: Dr. Dosik Hwang

Jan. 2016 - Feb. 2022

Research Intern, Philips Korea

Seoul, S.Korea

- Research Intern with Medical Artificial Intelligence Lab, Yonsei University

Oct. 2017 - Dec. 2017

Teaching Experience

Korea University

Seoul, S.Korea

Guest Lecturer

Mar. 2026

• The Fourth Industrial Revolution in Medical Science

- Presented a lecture on next-generation neuroimaging, including advanced quantitative and diffusion MRI driven by AI

Yonsei University

Guest Lecturer, Teaching Assistant

Seoul, S.Korea

Sep. 2021 - Dec. 2021

- **Introduction Artificial Intelligence**

- Presented a lecture on principles of deep learning and convolutional neural networks

Guest Lecturer, Teaching Assistant

Mar. 2021 - Jun. 2021

- **Medical Imaging Artificial Intelligence**

- Presented a lecture on MR image reconstruction using deep learning methods

Guest Lecturer, Teaching Assistant

Sep. 2020 - Dec. 2020

- **Medical Artificial Intelligence**

- Presented a lecture on principles of MRI and reconstruction methods for fast MRI

Teaching Assistant

Mar. 2018 - Jun. 2018

- **Introduction to Bioengineering for Electrical and Electronic Engineering**

Teaching Assistant

Mar. 2017 - Jun. 2017

- **Electrical and Electronic Engineering Capstone Design**

Grants

Advanced Neurovascular Imaging Using 7T MRI to Assess Long-Term Effects of Proton Radiotherapy in Pediatric Brain Tumor Survivors

National Institutes of Health (NIH),

National Cancer Institute (NCI)

Role: co-PI (Contact; Relinquished due to relocation), Total Direct Cost: \$2,433,015, Total Funding: \$3,473,277

Awarded Sep. 2026 (Project Period: 2026

- 2031)

- NIH R37CA309465 (Original PIs: **Y. Jun**, C. Jaimes)
- Successfully secured the grant as Contact PI, Relinquished the role prior to NOA due to a faculty appointment in South Korea

Rapid, Motion-Robust, and Low-Gadolinium MRI for Pediatric Brain Tumors

National Institutes of Health (NIH),

National Institute of Biomedical Imaging and Bioengineering (NIBIB)

Role: co-PI, Total Direct Cost: \$400,000, Total Funding: \$668,000

Sep. 2024 - Aug. 2027

- NIH R21EB036105 (PIs: **Y. Jun**, C. Jaimes)

Honors & Awards

INTERNATIONAL

2024 **ISMRM Junior Fellow**, The ISMRM 32nd Annual Meeting

Singapore

2024 **1st Place Winner, Best Oral Presentation**, The ISMRM 32nd Annual Meeting, Diffusion Study Group

Singapore

2024 **ISMRM Annual Meeting Program Committee (AMPC) Selected Abstract (Top 1%)**, The ISMRM 32nd Annual Meeting

Singapore

2024 **ISMRM Summa Cum Laude**, The ISMRM 32nd Annual Meeting

Singapore

2022-2023 **Distinguished Reviewer**, IEEE Transactions on Medical Imaging (IEEE TMI)

2023 **ISMRM Summa Cum Laude**, The ISMRM 31st Annual Meeting

Toronto, Canada

2021 **1st Rank**, Cross-Modality Domain Adaptation for Medical Image Segmentation (crossMoDA-2021 challenge)

Virtual Conference

2021 **ISMRM Magna Cum Laude (1)**, The ISMRM 29th Annual Meeting

Virtual Conference

2021 **ISMRM Magna Cum Laude (2)**, The ISMRM 29th Annual Meeting

Virtual Conference

2020 **3rd Rank**, fastMRI Challenge 2020, Facebook AI Research & NYU Langone Health

Virtual Conference

2020 **ISMRM Summa Cum Laude**, The ISMRM 28th Annual Meeting

Virtual Conference

2020 **ISMRM The Poster Award of 2nd Place (Silver)**, 2020 ISMRM Workshop on Data Sampling & Image Reconstruction

Sedona, US

2019 **4th Rank**, fastMRI Challenge 2019, Facebook AI Research & NYU Langone Health

Vancouver, Canada

2017 **ISMRM Summa Cum Laude**, The ISMRM 25th Annual Meeting

Hawaii, US

DOMESTIC

2026	MGB Radiology Innovation Award , Massachusetts General Brigham (MGB)	<i>Boston, US</i>
2021	Excellence Award , Medical Artificial Intelligence Datathon 2021, Ministry of Science and ICT and National Information Society Agency	<i>Seoul, S.Korea</i>
2021	Excellence Award , Hackathon of Development of AI-based Image Diagnosis using Medical Big Data 2021, Korea Testing Laboratory (KTL)	<i>Seoul, S.Korea</i>
2021	Best Paper Award , Graduate Student Paper Award, Yonsei University	<i>Seoul, S.Korea</i>
2019	Participation Prize , Samsung Humantech Paper Award (first author)	<i>Seoul, S.Korea</i>
2019	1st Rank and Grand Prize , HeLP Challenge 2018, Brain Tumor Segmentation Contest	<i>Seoul, S.Korea</i>
2018	Participation Prize , Samsung Humantech Paper Award (co-author)	<i>Seoul, S.Korea</i>
2017	Grand Prize , Yonsei Junior Convergence Science	<i>Seoul, S.Korea</i>

Scholarship

2023	ISMRM Trainee Stipend , ISMRM Workshop on Data Sampling and Image Reconstruction	<i>US</i>
2021	Dissertation Fellowship , Graduate Students Idea Incubation Fund, Yonsei University	<i>S.Korea</i>
2021	Academy Research Fellowship , Graduate Students Idea Incubation Fund, Yonsei University	<i>S.Korea</i>
2021	Best Paper Award Scholarship , Graduate Student Paper Award, Yonsei University	<i>S.Korea</i>
2020	ISMRM Trainee Stipend , ISMRM Workshop on Data Sampling and Image Reconstruction	<i>US</i>
2017-2019	ISMRM Educational Stipend , ISMRM	<i>US</i>
2019	Brain Korea 21 Plus Outstanding Student Fellow Scholarship , Korea Research Foundation	<i>S.Korea</i>
2018	Teaching Assistant Scholarship , Yonsei University	<i>S.Korea</i>
2017-2020	Brain Korea 21 Plus Scholarship , Korea Research Foundation	<i>S.Korea</i>
2016	Research Assistant Scholarship , Yonsei University	<i>S.Korea</i>
2012-2015	National Scholarship for Science & Engineering , Korea Student Aid Foundation	<i>S.Korea</i>

Invited Talks

From Quantitative to Diffusion MRI: Advanced Neuroimaging Driven by AI KU Graduate Program for Convergence & Translational Biomedicine • Korea University, Brain Convergence Research Center	<i>Seoul, S.Korea</i> <i>Aug. 2025</i>
AI Improvement in Image Quality and Analysis in Research and Clinical Practice ISMRM Pediatric MR Study Group Member-initiated Symposium • International Society for Magnetic Resonance in Medicine (ISMRM)	<i>Hawaii, US</i> <i>May. 2025</i>
Advanced neuroimaging using MRI: from quantitative MRI to diffusion MRI Brain Korea (BK) 21 Y-BASE R&E Institute • Yonsei University, School of Electrical and Electronic Engineering	<i>Seoul, S.Korea</i> <i>May. 2024</i>
Self-Supervised Learning for Rapid Quantitative MRI Athinoula A. Martinos Center for Biomedical Imaging • Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital	<i>Boston, US</i> <i>May. 2023</i>
Deep Model-based MR Parameter Mapping Network (DOPAMINE) for Fast MR Reconstruction 34th KSIIM Conference, 2020 • Korean Society of Imaging Informatics in Medicine	<i>Seoul, S.Korea</i> <i>Oct. 2020</i>
Medical Imaging Research using Artificial Intelligence HUFS AIM LAB, 2020 • The Catholic University of Korea, Eunpyeong St. Mary's Hospital	<i>Seoul, S.Korea</i> <i>Jan. 2020</i>

Presented Talks

Mesoscale Multiparametric qMRI at 7T with Multi-Echo vNav-QALAS and Self-Supervised Reconstruction ISMRM Annual Meeting, 2026 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2026	<i>Capetown, South Africa</i> <i>May. 2026</i>
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vNav-QALAS: Prospective Motion-Corrected 3D Multiparametric Mapping with Integrated Volumetric Navigators ISMRM Annual Meeting, 2026 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2026	<i>Capetown, South Africa</i> <i>May. 2026</i>
Efficient mesoscale multiparametric quantitative MRI using 3D-QALAS at 7T with self-supervised learning ISMRM Annual Meeting, 2025 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2025	<i>Hawaii, US</i> <i>May. 2025</i>
Phase Reversed Interleaved Multi-Echo (PRIME) with phase, field map and motion navigators for highly accelerated distortion-free diffusion MRI ISMRM Annual Meeting, 2025 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2025	<i>Hawaii, US</i> <i>May. 2025</i>
PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI ISMRM Annual Meeting, 2024 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2024	<i>Singapore</i> <i>May. 2024</i>
Rapid Pediatric Imaging with Zero-Shot Deep Subspace Reconstruction for Multiparametric Quantitative MRI ISMRM Annual Meeting, 2024 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2024	<i>Singapore</i> <i>May. 2024</i>
Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Multiparametric Quantitative MRI Using QALAS ISMRM Annual Meeting, 2023 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2023	<i>Toronto, Canada</i> <i>June. 2023</i>
Deep Subspace Reconstruction with Zero-Shot Learning for Multiparametric Quantitative MRI ISMRM Workshop on Data Sampling and Image Reconstruction, 2023 • International Society for Magnetic Resonance in Medicine (ISMRM) on Data Sampling and Image Reconstruction, 2023	<i>Sedona, US</i> <i>Jan. 2023</i>
Joint Reconstruction of MR Image and Coil Sensitivity Maps using Deep Model-based Network ISMRM Annual Meeting, 2021 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2021	<i>Virtual Conference</i> <i>May. 2021</i>
Deep Learning-based Automatic Detection and Segmentation of Brain Metastases Using Multi-Task Learning with 3D Black-Blood and GRE Imaging ISMRM Annual Meeting, 2021 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2021	<i>Virtual Conference</i> <i>May. 2021</i>
Deep Model-based MR Parameter Mapping Network (DOPAMINE) for Fast MR Reconstruction ISMRM Annual Meeting, 2020 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2020	<i>Virtual Conference</i> <i>Aug. 2020</i>
Deep Convolutional Neural Network for Acceleration of Magnetic Resonance Angiography (MRA) ISMRM Annual Meeting, 2017 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2017	<i>Hawaii, US</i> <i>Apr. 2017</i>

Publications - Preprints

Vendor-Agnostic Joint Relaxometry and Myelin Water Fraction Mapping with B1 and Motion Correction UD Gallastegi [†] , S Fujita [†] , Y Jun , AD Klauser, GF Piredda, T Hilbert, C Ariyurek, E Milshteyn, S Li, Y Chen, X Yong, KS Chan, Q Liu, S Yee, Y Rathi, M Zaitsev, JF Nielsen, O Afacan, C Jaimes, PE Grant, B Gagoski [#] , B Bilgic [#] • <i>arXiv preprint arXiv:2608.07687</i>	<i>2026</i>
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- MWF-MIMOSA for efficient simultaneous relaxometry and myelin water fraction mapping 2026
Y Chen, **Y Jun***, HG Shin, S Li, S Fujita, X Yong, J Kim, J Lee, GF Piredda, T Hilbert, A Bhatt, SY Huang, H Liu*, H Ye, S Nasr, B Gagoski, KS Chan, B Bilgic
• ***Co-corresponding Authors**, *arXiv preprint arXiv:2607.28984*
- Submillimeter postmortem and in vivo diffusion and susceptibility magnetic resonance imaging to characterize cortical micro-and meso-structures 2026
Y Uchida, HG Shin, LG Isaza, J Troncoso, A Bibic, J Gerold, D Sung, **Y Jun**, F Wang, Z Dong, S Huang, B Bilgic, PV Zijl, X Li, K Oishi
• *research square preprint rs-9612742/v1*
- Omni-QALAS: Optimized Multiparametric Imaging for Simultaneous T1, T2 and Myelin Water Mapping 2025
S Li, UD Gallastegi, S Fujita, Y Chen, P Xu, Y Choi, B Gagoski, H Ye, H Liu*, B Bilgic*, **Y Jun***
• ***Co-corresponding Authors**, *arXiv preprint arXiv:2510.13118*
- Explainable AI for Accelerated Microstructure Imaging: A SHAP-Guided Protocol on the Connectome 2.0 scanner 2025
Q Uhl, T Pavan, J Gerold, KS Chan, **Y Jun**, S Fujita, A Bhatt, Y Ma, Q Wang, HH Lee, SY Huang, B Bilgic, I Jelescu
• *arXiv preprint arXiv:2509.09513*
- MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI 2025
Y Chen, **Y Jun**, A Heydari, X Yong, J Kim, J Lee, H Liu, H Ye, B Gagoski, S Fujita*, B Bilgic*
• *arXiv preprint arXiv:2508.10184*
- Multishot Dual Polarity GRAPPA: Robust Nyquist Ghost Correction for multishot EPI 2025
Y Jiang, **Y Jun**, Q Liu, W Zhong, Y Rath, H Guo, B Bilgic
• *arXiv preprint arXiv:2507.18273*
- A Tutorial on MRI Reconstruction: From Modern Methods to Clinical Implications 2025
T Çukur, SUH Dar, VA Nezhad, **Y Jun**, TH Kim, S Fujita, B Bilgic
• *arXiv preprint arXiv:2507.16715*
- Evaluation of Synthetic Pediatric Brain MRI Using 3D-QALAS and Zero-DeepSub Reconstruction 2025
SF Ferracioli*, **Y Jun***, SAV Vasquez, VP Trujillo, H Griffin, S Fujita, E Milshteyn, B Bilgic, C Jaimes
• ***Co-first Authors**, *research square preprint rs-7025131/v1*
- PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI 2024
Y Jun*, Q Liu, T Gong, J Cho, S Fujita, X Yong, C Liao, ME Schmidt, S Nasr, C Jaimes, MS Gee, SY Huang, L Ning, A Yendiki, Y Rath, B Bilgic
• *arXiv preprint arXiv:2409.07375*
- NLCG-Net: A Model-Based Zero-Shot Learning Framework for Undersampled Quantitative MRI Reconstruction 2024
X Jian, **Y Jun**, J Cho, M Gao, X Yong, B Bilgic
• *arXiv preprint arXiv:2401.12004*
- Improved Multi-Shot Diffusion-Weighted MRI with Zero-Shot Self-Supervised Learning Reconstruction 2023
J Cho, **Y Jun**, X Wang, C Kobayashi, B Bilgic
• *arXiv preprint arXiv:2308.05103*

- Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Rapid Multiparametric Quantitative MRI Using 3D-QALAS 2023
Y Jun*, Y Arefeen, J Cho, S Fujita, X Wang, PE Grant, B Gagoski, C Jaimes, MS Gee[#], B Bilgic[#]
 • ***Corresponding Author**, *arXiv preprint arXiv:2307.01410*
- SDC-UDA: Volumetric Unsupervised Domain Adaptation Framework for Slice-Direction Continuous Cross-Modality Medical Image Segmentation 2023
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2305.11012*
- SSL-QALAS: Self-Supervised Learning for Rapid Multiparameter Estimation in Quantitative MRI Using 3D-QALAS 2023
Y Jun*, J Cho, X Wang, M Gee, PE Grant, B Bilgic[#], B Gagoski[#]
 • ***Corresponding Author**, *arXiv preprint arXiv:2302.14240*
- COSMOS: Cross-Modality Unsupervised Domain Adaptation for 3D Medical Image Segmentation based on Target-aware Domain Translation and Iterative Self-Training 2022
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2203.16557*
- Self-Training Based Unsupervised Cross-Modality Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation 2021
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2109.10674*
- Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction 2020
 MJ Muckley⁺, B Riemenschneider⁺, A Radmanesh, S Kim, G Jeong, J Ko, **Y Jun**, H Shin, D Hwang, M Mostapha, S Arberet, D Nickel, Z Ramzi, P Ciuciu, JL Starck, J Teuwen, D Karkalousos, C Zhang, A Sriram, Z Huang, N Yakubova, YW Lui, Florian Knoll
 • *arXiv preprint arXiv:2012.06318*

Publications - Peer-Review Journal

- MWF-MIMOSA for efficient simultaneous relaxometry and myelin water fraction mapping 2026
 Y Chen, **Y Jun***, HG Shin, S Li, S Fujita, X Yong, J Kim, J Lee, GF Piredda, T Hilbert, A Bhatt, SY Huang, H Liu^{*}, H Ye, S Nasr, B Gagoski, KS Chan, B Bilgic
 • ***Co-corresponding Authors**, *IEEE Transactions on Biomedical Engineering*, (under review)
- Vendor-Agnostic Joint Relaxometry and Myelin Water Fraction Mapping with B1 and Motion Correction 2026
 UD Gallastegi⁺, S Fujita⁺, **Y Jun**, AD Klauser, GF Piredda, T Hilbert, C Ariyurek, E Milshteyn, S Li, Y Chen, X Yong, KS Chan, Q Liu, S Yee, Y Rath, M Zaitsev, JF Nielsen, O Afacan, C Jaimes, PE Grant, B Gagoski[#], B Bilgic[#]
 • *Imaging Neuroscience*, (under review)
- Omni-QALAS: Optimized Multiparametric Imaging for Simultaneous T1, T2 and Myelin Water Mapping 2026
 S Li, UD Gallastegi, S Fujita, Y Chen, P Xu, KS Chan, Y Choi, J Kim, GF Piredda, T Hilbert, J Lee, H Ye, B Gagoski, H Liu^{*}, B Bilgic^{*}, **Y Jun***
 • ***Co-corresponding Authors**, *IEEE Transactions on Biomedical Engineering*

Explainable AI for Accelerated Microstructure Imaging: A SHAP-Guided Protocol on the Connectome 2.0 scanner Q Uhl, T Pavan, J Gerold, KS Chan, Y Jun, S Fujita, A Bhatt, Y Ma, Q Wang, HH Lee, SY Huang, B Bilgic[#], I Jelescu[#] • <i>Imaging Neuroscience</i>	2026
SeqEyes: An open-source toolbox for synchronized waveform/trajectory visualization in Pulseseq X Yong[*], Z Liu, Y Dong, H Hong, Y Chen, P Dubovan, Q Liu, J Kim, UD Gallastegi, Y Jun[*], B Bilgic • *Co-corresponding Authors, <i>Magnetic Resonance in Medicine</i>	2026
PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-corrected diffusion MRI Y Jun[*], Q Liu, T Gong, J Cho, S Fujita, X Yong, C Liao, ME Schmidt, S Nasr, C Jaimes, MS Gee, SY Huang, L Ning, A Yendiki, Y Rath, B Bilgic • *Corresponding Author, <i>Medical Image Analysis</i>, 111:104058	2026
Deep learning in fetal, infant, and toddler neuroimaging research JH Chin, MK Wyburd, V Ayzenberg, L Bayet, B Bilgic, EM Chen, Y Chen, A Dineen, S Fujita, J Liu, Y Jun, MC Camacho, L Zollei • <i>Developmental Cognitive Neuroscience</i>, 78:101680	2026
3D-QALAS synthetic MRI with Zero-DeepSub in children: initial experience including post-contrast imaging feasibility SF Ferracioli^{†*}, Y Jun^{†*}, SAV Vasquez, VP Trujillo, H Griffin, S Fujita, E Milshteyn, B Bilgic, C Jaimes • † Co-first Authors, *Co-corresponding Authors, <i>Pediatric Radiology</i>, 56(4):867-877	2026
Multishot Dual Polarity GRAPPA: Robust Nyquist Ghost Correction for multishot EPI Y Jiang, Y Jun, Q Liu, W Zhong, Y Rath, H Guo, B Bilgic • <i>Magnetic Resonance in Medicine</i>, 95(5):2726-2736	2026
MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI Y Chen, Y Jun, A Heydari, X Yong, J Kim, J Lee, H Liu, H Ye, B Gagoski, S Fujita[#], B Bilgic[#] • <i>Magnetic Resonance in Medicine</i>, 95(3):1528-1544	2026
A Tutorial on MRI Reconstruction: From Modern Methods to Clinical Implications T Çukur, SUH Dar, VA Nezhad, Y Jun, TH Kim, S Fujita, B Bilgic • <i>IEEE Transactions on Biomedical Engineering</i>	2025
Vendor-agnostic 3D multiparametric relaxometry improves cross-platform reproducibility S Fujita, B Gagoski, JF Nielsen, M Zaitsev, Y Jun, J Cho, X Yong, Q Uhl, P Xu, E Milshteyn, S Imam, Q Liu, Q Chen, O Afacan, JE Kirsch, Y Rath, B Bilgic • <i>Magnetic Resonance in Medicine</i>, 94(3):937-948	2025
Beyond the Conventional Structural MRI: Clinical Application of Deep Learning Image Reconstruction and Synthetic MRI of the Brain Y Choi, JS Ko, JE Park, G Jeong, M Seo, Y Jun, S Fujita, B Bilgic • <i>Investigative Radiology</i>, 60(1):27-42	2025
Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Rapid Multiparametric Quantitative MRI Using 3D-QALAS Y Jun[*], Y Arefeen, J Cho, S Fujita, X Wang, PE Grant, B Gagoski, C Jaimes, MS Gee[#], B Bilgic[#] • *Corresponding Author, <i>Magnetic Resonance in Medicine</i>, 91(6):2459-2482	2024

- SSL-QALAS: Self-Supervised Learning for Rapid Multiparameter Estimation in Quantitative MRI Using 3D-QALAS 2023
Y Jun⁺, J Cho, X Wang, M Gee, PE Grant, B Bilgic[#], B Gagoski[#]
 • ⁺ **Corresponding Author**, *Magnetic Resonance in Medicine*, 90(5):2019-2032
- Deep learning referral suggestion and tumour discrimination using explainable artificial intelligence applied to multiparametric MRI 2023
 H Shin, JE Park, **Y Jun**, T Eo, J Lee, JE Kim, DH Lee, HH Moon, SI Park, S Kim, D Hwang, HS Kim
 • *European Radiology*, 33:5859–5870
- Intelligent Noninvasive Meningioma Grading with a Fully Automatic Segmentation using Interpretable Multiparametric Deep Learning 2023
Y Jun⁺, YW Park⁺, H Shin⁺, Y Shin, JR Lee, K Han, SS Ahn, SM Lim, D Hwang, SK Lee
 • ⁺ **Co-first Authors**, *European Radiology*, 33(9):6124-6133
- Ultrathin crystalline-silicon-based strain gauges with deep learning algorithms for silent speech interfaces 2022
 T Kim⁺, Y Shin⁺, K Kang⁺, K Kim⁺, G Kim⁺, Y Byeon⁺, H Kim, Y Gao, JR Lee, G Son, T Kim, **Y Jun**, J Kim, J Lee, S Um, Y Kwon, BG Son, M Cho, M Sang, J Shin, K Kim, J Suh, H Choi, S Hong, H Cheng, HG Kang^{*}, D Hwang^{*}, KJ Yu^{*}
 • *Nature Communications*, 13:5815
- Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction 2021
 MJ Muckley⁺, B Riemschneider⁺, A Radmanesh, S Kim, G Jeong, J Ko, **Y Jun**, H Shin, D Hwang, M Mostapha, S Arberet, D Nickel, Z Ramzi, P Ciuciu, JL Starck, J Teuwen, D Karkalousos, C Zhang, A Sriram, Z Huang, N Yakubova, YW Lui, Florian Knoll
 • *IEEE Transactions on Medical Imaging*, 40(9):2306-2317
- Deep model-based magnetic resonance parameter mapping network (DOPAMINE) for fast T1 mapping using variable flip angle method 2021
Y Jun, H Shin, T Eo, T Kim, D Hwang
 • *Medical Image Analysis*, 70:102017
- Robust performance of deep learning for automatic detection and segmentation of brain metastases using three-dimensional black-blood and three-dimensional gradient echo imaging 2021
 YW Park⁺, **Y Jun**⁺, Y Lee, K Han, C An, SS Ahn, D Hwang, SK Lee
 • ⁺ **Co-first Authors**, *European Radiology*, 31:6686-6695
- The Latest Trends in Attention Mechanisms and Their Application in Medical Imaging 2020
 H Shin, J Lee, T Eo, **Y Jun**, S Kim, D Hwang
 • *Journal of the Korean Society of Radiology*, 81(6):1305-1333
- Accelerating Cartesian MRI by domain-transform manifold learning in phase-encoding direction 2020
 T Eo⁺, H Shin⁺, **Y Jun**, T Kim, D Hwang
 • *Medical Image Analysis*, 63:101689
- Parallel imaging in time-of-flight magnetic resonance angiography using deep multistream convolutional neural networks 2019
Y Jun, T Eo, H Shin, T Kim, HJ Lee, D Hwang
 • *Magnetic Resonance in Medicine*, 81(6):3840-3853

- Megahertz-wave-transmitting conducting polymer electrode for device-to-device integration 2019
 T Kim, G Kim, H Kim, HJ Yoon, T Kim, **Y Jun**, TH Shin, S Kang, J Cheon, D Hwang, BW Min, W Shim
 • *Nature Communications*, 10:653
- Deep-learned 3D black-blood imaging using automatic labelling technique and 3D convolutional neural networks for detecting metastatic brain tumors 2018
Y Jun, T Eo, T Kim, H Shin, D Hwang, SH Bae, YW Park, HJ Lee, BW Choi, SS Ahn
 • *Scientific Reports*, 8:9450
- KIKI-net: cross-domain convolutional neural networks for reconstructing undersampled magnetic resonance images 2018
 T Eo, **Y Jun**, T Kim, J Jang, HJ Lee, D Hwang
 • *Magnetic Resonance in Medicine*, 80(5):2188-2201
- High-SNR multiple T2 (*)-contrast magnetic resonance imaging using a robust denoising method based on tissue characteristics 2017
 T Eo, T Kim, **Y Jun**, H Lee, SS Ahn, DH Kim, D Hwang
 • *Journal of Magnetic Resonance Imaging*, 45(6):1835-1845

Publications - Conference Papers

- Improved Multi-Shot Diffusion-Weighted MRI with Zero-Shot Self-Supervised Learning Reconstruction 2023
 J Cho, **Y Jun**, X Wang, C Kobayashi, B Bilgic
 • *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp.457-466
- SDC-UDA: Volumetric Unsupervised Domain Adaptation Framework for Slice-Direction Continuous Cross-Modality Medical Image Segmentation 2023
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp.7412-7421
- Evaluation of the Robustness of Learned MR Image Reconstruction to Systematic Deviations Between Training and Test Data for the Models from the fastMRI Challenge 2021
 PM Johnson, G Jeong, K Hammernik, J Schlemper, C Qin, J Duan, D Rueckert, J Lee, N Pezzoti, ED Weerdt, S Yousefi, MS Elmahdy, JHFV Gemert, C Schuelke, M Doneva, T Nielsen, S Kastrulin, BPF Lelieveldt, MJPV Osch, M Staring, EZ Chen, P Wang, X Chen, T Chen, VM Patel, S Sun, H Shin, **Y Jun**, T Eo, S Kim, T Kim, D Hwang, P Putzky, D Karkalousos, J Teuwen, N Miriakov, B Bakker, M Caan, M Welling, MJ Muckley, F Knoll
 • *International Workshop on Machine Learning for Medical Image Reconstruction (MLMIR)*, pp. 25-34
- Joint Deep Model-based MR Image and Coil Sensitivity Reconstruction Network (Joint-ICNet) for Fast MRI 2021
Y Jun, H Shin, T Eo, D Hwang
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Publications - Conference Abstracts

- Mesoscale Multiparametric qMRI at 7T with Multi-Echo vNav-QALAS and Self-Supervised Reconstruction 2026
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- vNav-QALAS: Prospective Motion-Corrected 3D Multiparametric Mapping with Integrated Volumetric Navigators 2026
Y Jun, Y Chen, X Yong, S Yee, R Frost, AVD Kouwe, MS Gee, PE Grant, B Gagoski, B Bilgic, C Jaimes
• [***Oral Presentation**] *International Society for Magnetic Resonance in Medicine (ISMRM)*
- Omni-QALAS: Optimized Multiparametric Imaging for Simultaneous T1, T2 and Myelin Water Mapping 2026
S Li, UD Gallastegi, S Fujita, Y Chen, P Xu, KS Chan, B Gagoski, Y Choi, J Kim, GF Piredda, T Hilbert, J Lee, H Ye, H Liu, B Bilgic, Y Jun
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- MWF-MIMOSA for Highly-efficient Joint Relaxometry and Myelin Water Fraction Mapping 2026
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- Multi-Compartment Relaxometry for myelin water imaging with magnetic susceptibility source separation (MCR-MWI-Chisep) 2026
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- Predicting Full-Dose Gadolinium T1 and T2 Maps from Low-Dose Multiparametric Acquisitions 2026
UD Gallastegi, Y Jun, Y Chen, X Yong, C Calixto, S Valencia, S Fujita, PE Grant, S Yee, J Stockmann, B Gagoski, C Jaimes, B Bilgic
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A Vurankaya, Y Jun , J Cho, B Bilgic	
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• <i>International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on Data Sampling and Image Reconstruction</i>	
Model-Based Phase-Difference Reconstruction for Accelerated Phase-Based T2 Mapping	2023
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• <i>International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on Data Sampling and Image Reconstruction</i>	
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- Joint Generation of Multi-contrast Magnetic Resonance Images and Segmentation Map Using StyleGAN2-based Generative Network 2022
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- Arbitrary Missing Contrast Generation Using Multi-Contrast Generative Network with An Encoder Network 2022
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H Shin, JE Park, **Y Jun**, HS Kim, D Hwang
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- Parallel Imaging based on k-x Domain Interpolation using Deep Neural Networks 2019
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• *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp. 4660
- Deep-learned 3D black-blood imaging using automatic labeling technique and 3D convolutional neural networks for detection of metastatic brain tumors 2018
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- Brain Vessel Extraction without MRA / V using Deep Convolutional Neural Network 2018
H Shin, **Y Jun**, T Kim, T Eo, S Ahn, D Hwang
• *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp. 3171
- Automatic Selection of Optimal Regularization Parameters in Compressed Sensing using No Reference Magnetic Resonance Image Quality Assessment 2018
K Bang, J Jang, **Y Jun**, H Jang, H Lee, D Hwang
• *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp. 2816

- Deep Sinogram Learning for Radial MRI: Comparison with k-space and Image Learning 2018
T Kim, T Eo, D Park, **Y Jun**, D Hwang
• *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp. 2799
- Reconstruction of brain vessel signals from undersampled time-of-flight magnetic resonance angiography using deep learning 2018
Y Jun, T Eo, H Shin, T Kim, HJ Lee, H Jang, D Hwang
• *The 21th Annual Meeting of the the Korean Society for Brain and Neural Sciences (KSBNS)*, pp. 1097
- Deep Convolutional Neural Network for Acceleration of Magnetic Resonance Angiography (MRA) 2017
Y Jun, T Eo, T Kim, J Jang, D Hwang
• [***Oral Presentation**] [***Summa Cum Laude**] *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp. 0686
- Cascaded Convolutional Neural Network (CNN) for Reconstruction of Undersampled Magnetic Resonance (MR) Images 2017
T Eo, **Y Jun**, T Kim, J Jang, D Hwang
• [***Summa Cum Laude**] *International Society for Magnetic Resonance in Medicine (ISMRM)* pp. 3974

Patents

- Phase Reversed Interleaved Multi-Echo (PRIME) acquisition for highly accelerated distortion-free diffusion MRI 2025
B Bilgic, J Cho, S Fujita, **Y Jun**, Q Liu, Y Rathi, X Yong
• *Provisional #, US63/803,522, (US)*
- Physics-informed approach to reduce dosage of gadolinium-based contrast agents for magnetic resonance imaging exams 2025
B Bilgic, J Cho, S Fujita, MS Gee, C Jaimes, **Y Jun**, J Stockmann, X Wang
• *Provisional #, US63/803,462, (US)*
- MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI 2025
B Bilgic, Y Chen, S Fujita, B Gagoski, **Y Jun**
• *Provisional #, US63/803,492, (US)*
- Method And Device For Correcting Medical Image Using Phantom 2022
D Hwang, **Y Jun**, T Kim, Y Lee
• *Registered, 10-2481027, (S. Korea)*
- Apparatus And Method For Reconstructing MR Parameter Map 2022
D Hwang, **Y Jun**, H Shin, T Eo, T Kim
• *Registered, 10-2352004, (S. Korea)*
- Device And Method For Reconstructing Magnetic Resonance Image Thereof 2021
D Hwang, T Eo, T Kim, **Y Jun**
• *Registered, 10-2233996, (S. Korea)*
- Learning Apparatus and Method for Generating Encephaloma Discriminative Image, Apparatus and Method for Generating Encephaloma Discriminative Image, and Recording Medium thereof 2018
SS Ahn, T Eo, **Y Jun**, D Hwang, BW Choi, T Kim, HJ Lee
• *Registered, 10-1928213, (S. Korea)*
- Device and Method for Reconstructing Undersampled Magnetic Resonance Image 2018
D Hwang, T Eo, **Y Jun**, T Kim
• *Registered, 10-1886575, (S. Korea)*

Activities

Ad Hoc Reviewer

- IEEE Transactions on Medical Imaging (IEEE TMI) (*Distinguished Reviewer)
- Magnetic Resonance in Medicine
- Medical Physics
- Artificial Intelligence in Medicine
- Communications Biology
- Scientific Reports
- Imaging Neuroscience
- Neuroradiology
- Brain Informatics
- Engineering Applications of Artificial Intelligence
- Frontiers in Pediatrics
- Quantitative Imaging in Medicine and Surgery
- Biomedical Engineering Letters
- Journal of Neuro-Oncology
- BMC Medical Imaging
- BMC Cancer
- BMC Nephrology
- IEEE Journal of Biomedical and Health Informatics (IEEE JBHI)
- IEEE Access
- IEEE Sensors Letters
- International Society for Magnetic Resonance in Medicine (ISMRM 2022-2026)
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2020-2025)
- European Conference on Computer Vision (ECCV 2024)
- International Conference on Computer Vision (ICCV 2025)
- Conference on Computer Vision and Pattern Recognition (CVPR 2025-2026)
- Neural Information Processing Systems (NeurIPS 2026)

Conference Moderator

- Moderator of International Society for Magnetic Resonance in Medicine (ISMRM 2024-2025)
- Poster Facilitator of International Society for Magnetic Resonance in Medicine (ISMRM 2021)

Professional Societies

- Full Member of International Society for Magnetic Resonance in Medicine (ISMRM) (2025-)
- Trainee Member of International Society for Magnetic Resonance in Medicine (ISMRM) (2017-2024)
- ISMRM Study Groups (Quantitative MRI, Ultra-high Field MR, Diffusion, Pediatric MR)

Skills

Programming	Python, Matlab, Pytorch, Tensorflow/Keras, C/C++
MRI Sequence Design	Pulseseq
Languages	Korean, English

References

Available upon request